

001 / 100 **LEGENDARY** 1982

BLIND SIGNATURES



UNTRACEABLE PAYMENTS.

THE HIDDEN MESSAGE

001 / BLIND SIGNATURES / DAVID CHAUM

Imagine taking a letter to someone whose signature you need. You want them to confirm it is valid, but you do not want them to read what is inside. In LORE's illustration, the envelope stays closed as the stamper lifts away. That gap between the signer and the hidden message is the idea behind a blind signature.

In 1982, David Chaum presented Blind Signatures for Untraceable Payments at CRYPTO '82. The requester disguises a message; a signer signs the disguised version; and the requester removes the disguise. The result can still be verified against the original message, even though the signer did not see it at signing time.

Chaum applied the idea to electronic payments. A bank could sign a digital payment token at withdrawal, then recognise a valid token when it was spent. Because the bank signed a blinded version, it could not simply match that later token to the particular one it saw at withdrawal. The card's line, UNTRACEABLE PAYMENTS., expresses the aim of the proposed scheme. It does not mean every surrounding detail of a real payment is anonymous.

The card freezes the instant after the seal is made. An old-fashioned envelope turns an abstract cryptographic move into a scene you can read at a glance.

HIDDEN IN THE ART

01 THE FRESH SEAL

The raised stamper and its shadow show the mark has just been made. The envelope stands for a message authenticated without being read.

02 THE CONCEALED SLIP

The partly hidden payment slip points to the payment application: withdrawal and later spending need not be directly linked.

03 FADING FOOTPRINTS

The trail breaks across the desk, a metaphor for the intended break in the transaction link. Other information can still compromise privacy.

04 THE DRAWER DIAGRAM

The node sketch is a deliberate forward reference to the 2008 Bitcoin whitepaper, not a real object from Chaum's 1982 setting.

SOURCE & ART NOTE

David Chaum, Blind Signatures for Untraceable Payments, Proceedings of CRYPTO '82, pp. 199-203. Chaum's publication list records 1982. The room, computer, envelope, drawer and characters are illustrative, not a reconstruction of his workspace.

Original paper: chaum.com/wp-content/uploads/2022/01/Chaum-blind-signatures.pdf